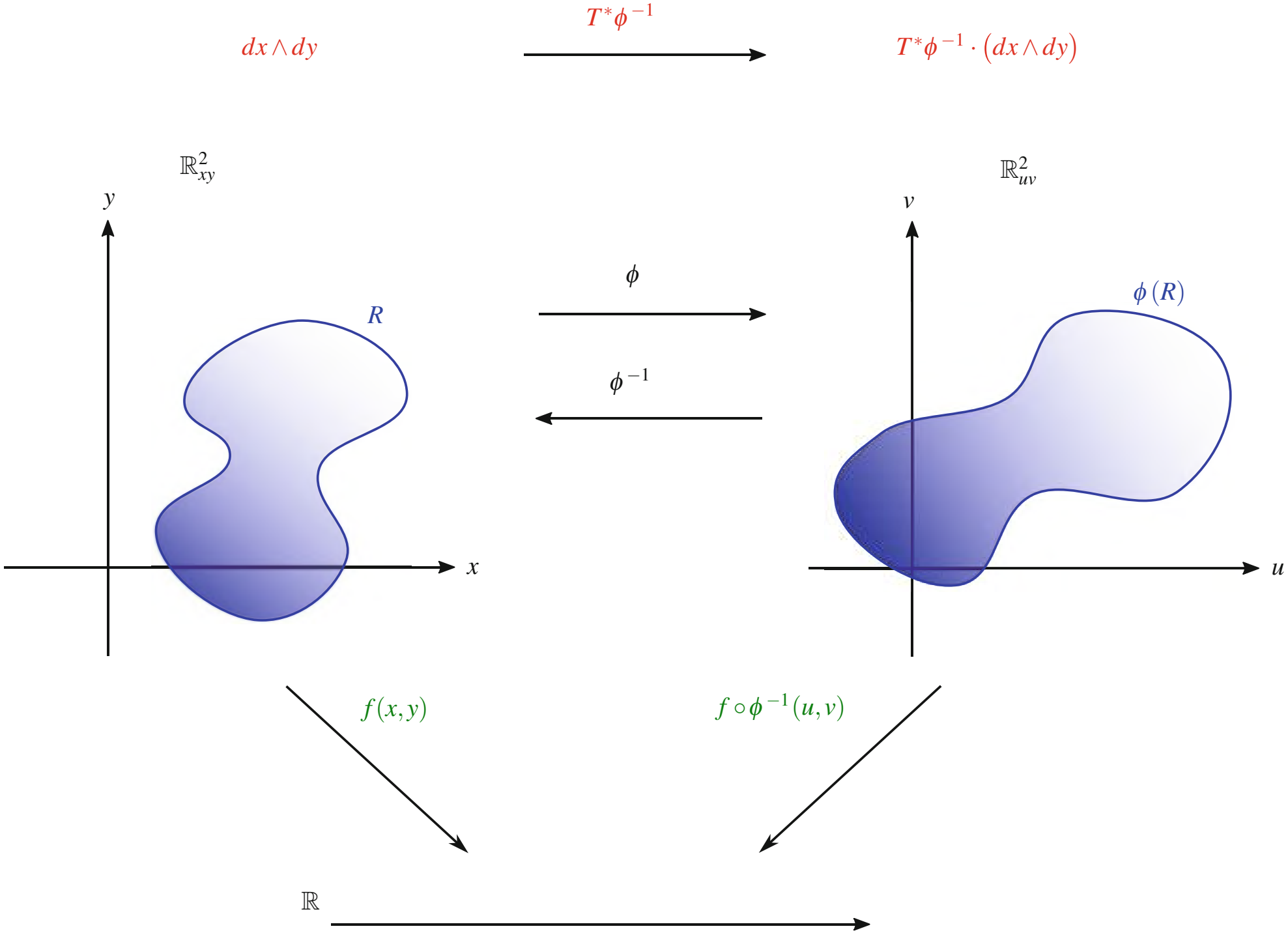

$dx \wedge dy$
$T^*\phi^{-1}$
$T^*\phi^{-1} \cdot (dx \wedge dy)$
$\mathbb{R}^2_{xy}$
y
x
R
ϕ
ϕ^{-1}
$\mathbb{R}^2_{uv}$
v
u
$\phi(R)$
$f(x,y)$
$f \circ \phi^{-1}(u,v)$
$\mathbb{R}$